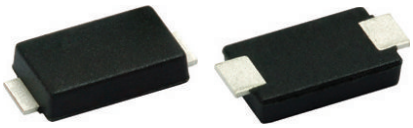
Hyperfast Rectifier, 1 A FRED Pt®

eSMP® Series



Top View Bottom View

SlimSMA HV (DO-221AC)



LINKS TO ADDITIONAL RESOURCES

3D Models

PRIMARY CHARACTERISTICS	
$I_{F(AV)}$	1 A
V_R	1200 V
V_F at I_F	1.45 V
t_{rr}	50 ns
T_J max.	175 °C
Package	SlimSMA HV (DO-221AC)
Circuit configuration	Single

FEATURES

- Minimum creepage distance 3.2 mm guaranteed by design
- Comparative Tracking Index: CTI ≥ 600
- Hyperfast recovery time, reduced Q_{rr}, and soft recovery
- 175 °C maximum operating junction temperature
- Specified for output and snubber operation
- Low forward voltage drop
- Low leakage current
- Meets MSL level 1, per J-STD-020, LF maximum peak of 260 °C
- Material categorization: for definitions of compliance please see www.vishay.com/doc?99912

RoHS COMPLIANT
HALOGEN FREE

DESCRIPTION / APPLICATIONS

State of the art hyperfast recovery rectifiers specifically designed with optimized performance of forward voltage drop and hyperfast recovery time.

The planar structure and the platinum doped life time control guarantee the best overall performance, ruggedness, and reliability characteristics.

These devices are intended for use as clamp, snubber and freewheeling diode in a flyback aux power supplies, bootstrap and desaturate for HV MOSFET and IGBT driver, high frequency rectifiers in a cuk and sepic circuit for LED lighting.

Their extremely optimized stored charge and low recovery current minimize the switching losses and reduce power dissipation in the switching element.

MECHANICAL DATA

Case: SlimSMA HV (DO-221AC)

Molding compound meets UL 94 V-0 flammability rating

Terminals: matte tin plated leads, solderable per J-STD-002

Polarity: color band denotes cathode end

ABSOLUTE MAXIMUM RATINGS				
PARAMETER	SYMBOL	TEST CONDITIONS	VALUES	UNITS
Peak repetitive reverse voltage	V_{RRM}		1200	V
Average rectified forward current	$I_{F(AV)}$	T_{sp} = 139 °C, DC conduction	1	A
Non-repetitive peak surge current	I_{FSM}	T_J = 25 °C, 8.3 ms sine pulse	14	
Operating junction and storage temperatures	T_J, T_{Stg}		-55 to +175	°C

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ELECTRICAL SPECIFICATIONS (T_J = 25 °C unless otherwise specified)						
PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNITS
Breakdown voltage, blocking voltage	V_{BR}, V_R	I_R = 100 µA	1200	-	-	V
Forward voltage drop	V_F	I_F = 1 A	-	1.85	2.30	V
		I_F = 1 A, T_J = 125 °C	-	1.55	1.75	V
		I_F = 1 A, T_J = 150 °C	-	1.45	1.65	V
Reverse leakage current	I_R	V_R = V_R rated	-	-	2	µA
		T_J = 125 °C, V_R = V_R rated	-	-	20	µA
Junction capacitance	C_T	V_R = 1200 V, 1 MHz	-	2.5	-	pF
		V_R = 4 V, 1 MHz	-	5.5	-	pF

DYNAMIC RECOVERY CHARACTERISTICS (T_J = 25 °C unless otherwise specified)							
PARAMETER	SYMBOL	TEST CONDITIONS		MIN.	TYP.	MAX.	UNITS
Reverse recovery time	t_{rr}	I_F = 0.5 A, I_R = 1 A, I_{rr} = 0.25 A		-	40	50	ns
		T_J = 25 °C	I_F = 1 A, dI_F/dt = 200 A/µs, V_R = 800 V	-	91	-	ns
		T_J = 125 °C		-	120	-	ns
Peak recovery current	I_{RRM}	T_J = 25 °C		-	3.0	-	A
		T_J = 125 °C		-	4.0	-	A
Reverse recovery charge	Q_{rr}	T_J = 25 °C		-	105	-	nC
		T_J = 125 °C		-	200	-	nC

THERMAL AND MECHANICAL SPECIFICATIONS						
PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNITS
Maximum junction and storage temperature range	T_J, T_{Stg}		-55	-	175	°C
Thermal resistance, junction to mount	R_{thJM} (1)	Device mounted on PCB with 2 x 3.5 mm soldering lands	-	20	23	°C/W
Thermal resistance, junction to ambient	R_{thJA}	Device mounted on PCB with recommended pad size	-	120	-	°C/W
Approximate weight			0.032			g
Marking device		Case style SlimSMA HV (DO-221AC)	1X12			

Note

(1) Thermal resistance junction to mount follows JEDEC® 51-14 transient dual interface test method (TDIM)

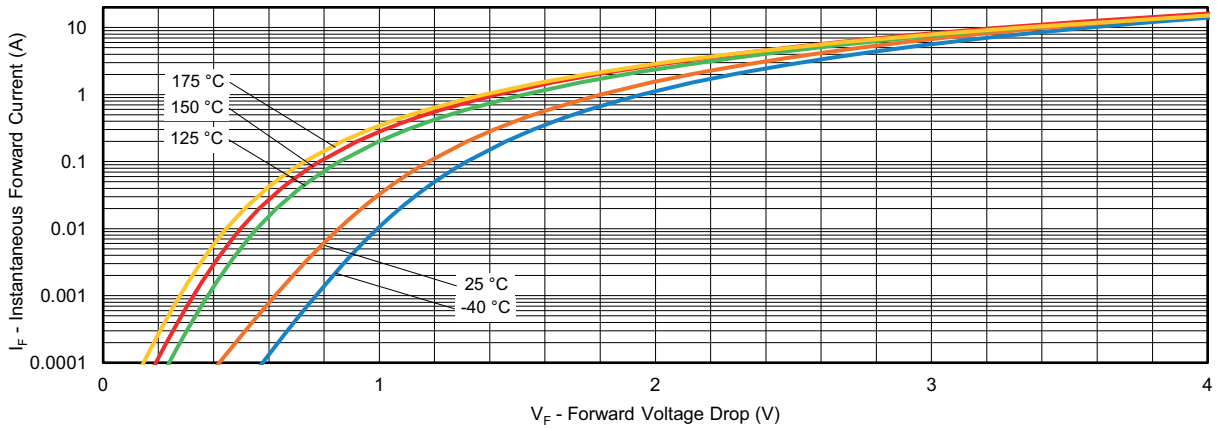


Fig. 1 - Typical Forward Voltage Drop Characteristics

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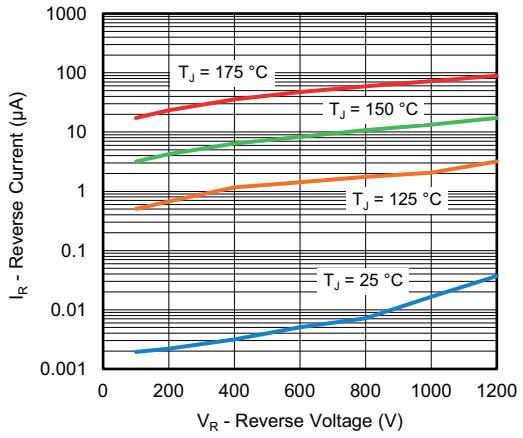


Fig. 2 - Typical Values of Reverse Current vs. Reverse Voltage

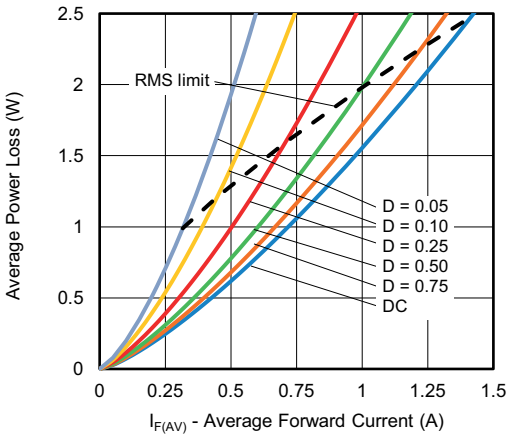


Fig. 5 - Forward Power Loss Characteristics

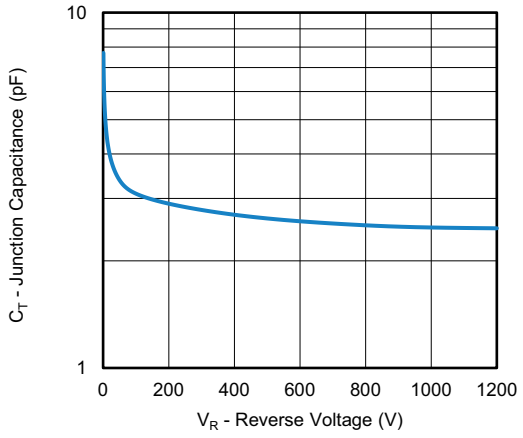


Fig. 3 - Typical Junction Capacitance vs. Reverse Voltage

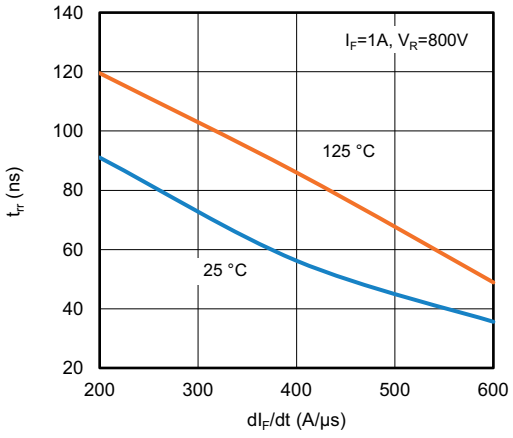


Fig. 6 - Typical Reverse Recovery Time vs. dI_F/dt

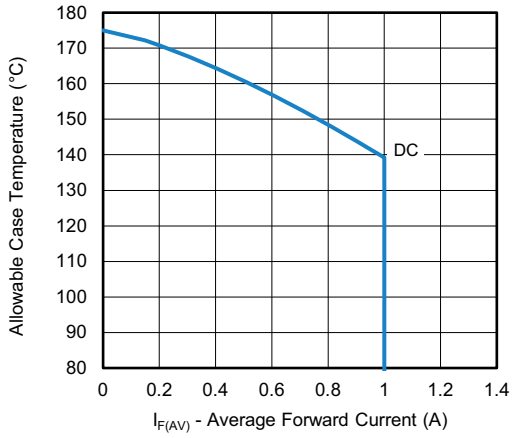


Fig. 4 - Maximum Allowable Case Temperature vs. Average Forward Current

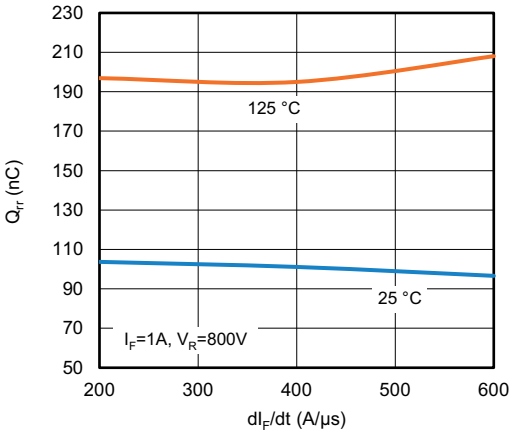


Fig. 7 - Typical Stored Charge vs. dI_F/dt

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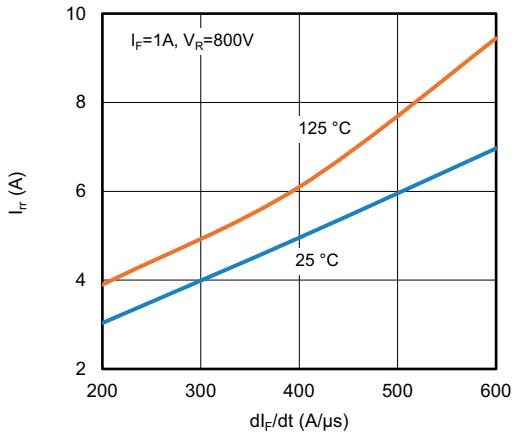


Fig. 8 - I_{rr} (A) vs. dI_F/dt

Note

(1) Formula used: $T_C = T_J - (Pd + Pd_{REV}) \times R_{thJC}$;
Pd = forward power loss = $I_{F(AV)} \times V_{FM}$ at ($I_{F(AV)}$/D) (see fig. 5);
Pd_{REV} = inverse power loss = $V_{R1} \times I_R$ (1 - D); I_R at V_{R1} = rated V_R

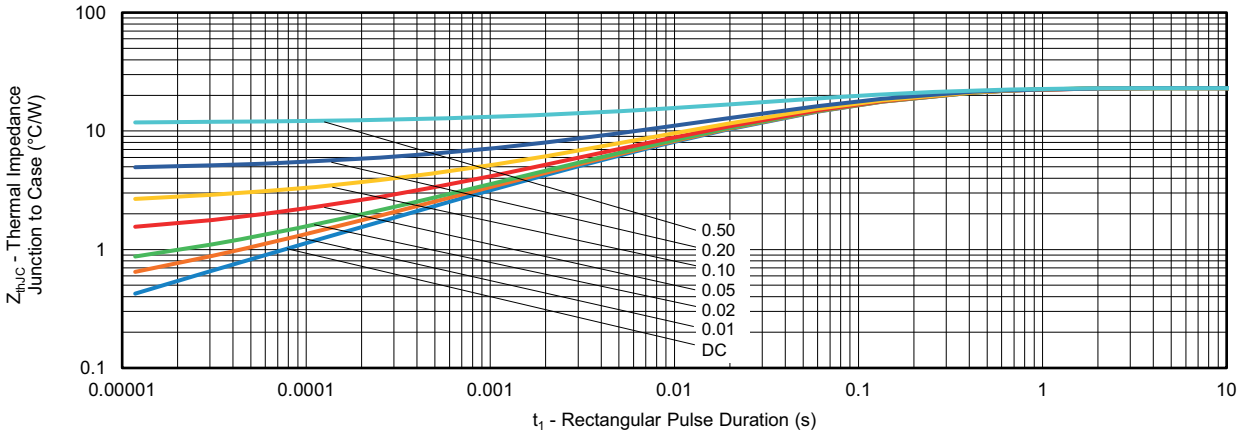


Fig. 9 - Transient Thermal Impedance, Junction to Case

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ORDERING INFORMATION TABLE

Device code

VS-	E	7	J	X	01	12	-M3
1	2	3	4	5	6	7	8

1 - Vishay Semiconductors product

2 - Circuit configuration:
E = single diode

3 - 7 = FRED generation 7

4 - J = SlimSMA package

5 - Process type,
X = hyperfast recovery

6 - Current rating (01 = 1 A)

7 - Voltage code (12 = 1200 V)

8 - M3 = halogen-free, RoHS-compliant, and terminations lead (Pb)-free

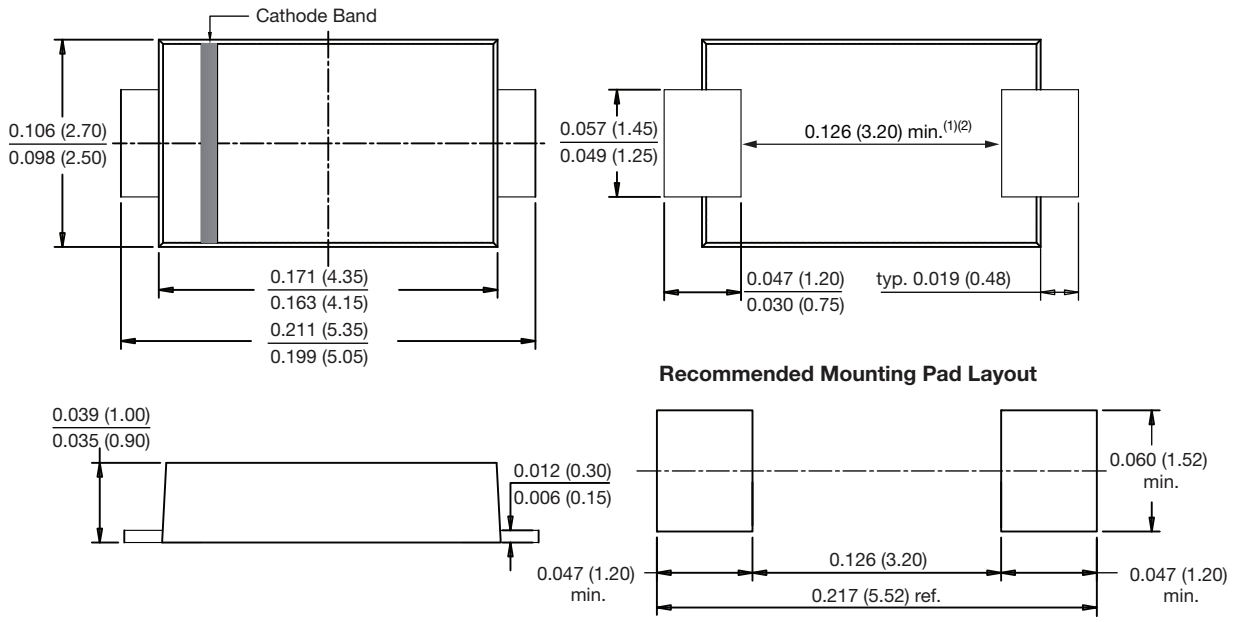
ORDERING INFORMATION (Example)

PREFERRED P/N	QUANTITY PER REEL	BASE QUANTITY	PACKAGING DESCRIPTION
VS-E7JX0112-M3/H	3500	3500	7"diameter plastic tape and reel
VS-E7JX0112-M3/I	14 000	14 000	13"diameter plastic tape and reel

LINKS TO RELATED DOCUMENTS	
Dimensions	www.vishay.com/doc?97278
Part marking information	www.vishay.com/doc?95562
Packaging information	www.vishay.com/doc?88869
SPICE model	www.vishay.com/doc?97550

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DIMENSIONS in inches (millimeters)



Notes

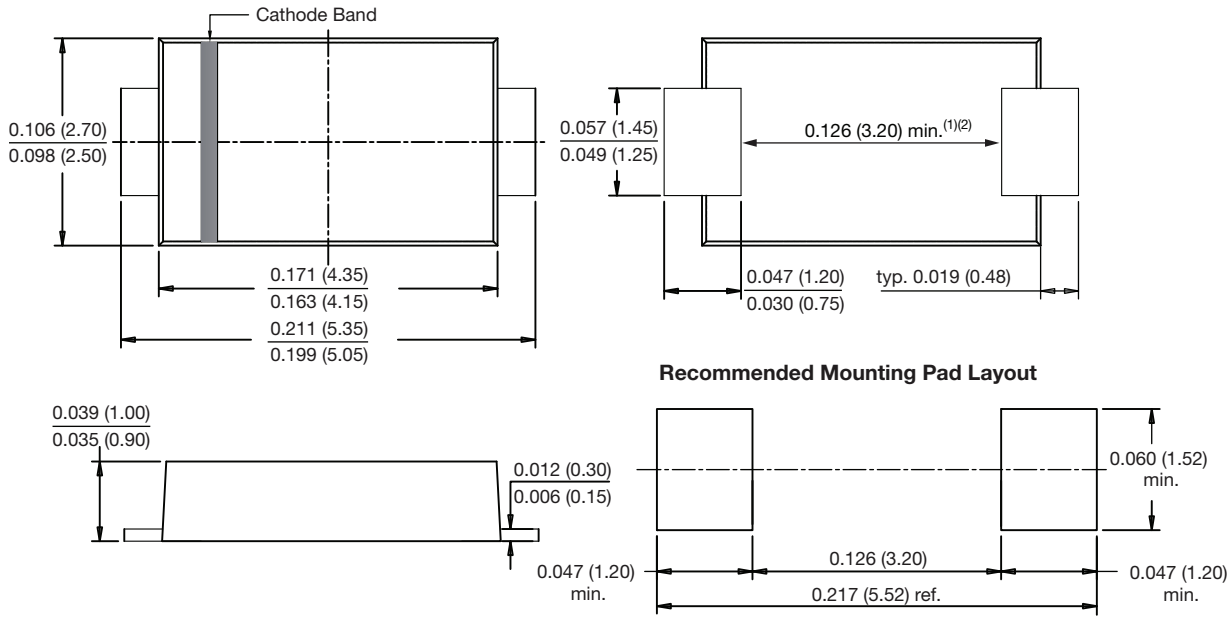
(1) Minimum creepage distance is defined and guaranteed by design

(2) For high voltage applications, end users should consider the relevant guidelines and normative on creepage and clearance distances between device terminals and PCB pads.

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SlimSMA HV (DO-221AC)

DIMENSIONS in inches (millimeters)



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